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Research Topic: Anthropogenic Heavy Metal Contamination and Bioaccumulation in *Channa striata* Populations: Implications for Aquaculture in the Ayeyarwady Region, Myanmar

Anthropogenic Heavy Metal Contamination and Bioaccumulation in *Channa striata* Populations: Implications for Aquaculture in the Ayeyarwady Region, Myanmar

Introduction

As the principal, largest and the most significant commercial waterway in Myanmar, the Ayeyarwady is considered as a 'lifeline' in factors of national economic stability, public sustenance and local livelihoods – particularly in subsistence fisheries. Among industries that rely on the Ayeyarwady, the aquaculture and aquaponics industries demonstrate the most prominent dependency, with the Ayeyarwady housing approximately 90% of Myanmar's freshwater aquaculture fisheries, which contributes to around 2% of national GDP [1]. Nonetheless, the rapid propagation of industrial zones inevitably compromises proximate water quality, as the waste produced by the mass production of goods are dumped into these mediums.

Despite major economic dependency on the Ayeyarwady region, evident and empirical data on the impacts of expanding industrial zones on commercial fish species, such as *Channa striata* – the striped snakehead, remains bounded in terms of research. By exploring the bioaccumulation of heavy metals (such as Pb, Cd, As, Hg and Cr) and how they transfer from contaminated waters to the gills and livers of commercial fish species, primarily *C. striata*, this research targets to fill in these gaps and further mitigate problems concerning ecological and economic baselines.

Literature Review

The Ayeyarwady region presents as an essential factor to economic and ecological factors, housing ecosystems that are increasingly vulnerable to anthropogenic pressure due to urbanization and industrialization of the region [3]. These anthropogenic progressions have also played a major role in the escalation of xenobiotics in these regions, making water sample analysis a critical research topic [9]. Geospatial mapping of the Ayeyarwady Delta has highlighted the need to study ecological and public health hazards regarding the region [9].

As a benthic carnivore, *C. striata* is vulnerable to bioaccumulation due to biomagnification through the food chain/web. Heavy metal absorption from water to *C. striata* species is demonstrated to be tied to the low-feeding behavior of the species [2]. Research on cadmium and lead concentrations among related species of snakeheads highlights that accumulations in the muscle tissues act as a sensitive biomarker for physiological stress, altering condition factor (K) and metabolic rates [7][8]. Studies suggest that environmental variables such as pH fluctuations accelerate heavy metal ion absorption through the fish gill membranes [9].

As shown in the following figure, Figure 1, the species of *C. striata* is assessed as a dominant member in terms of population in the region, with annual catch of the species exceeding 35,000 tons [3], accentuating higher chances of adverse heavy metal impacts upon these species.

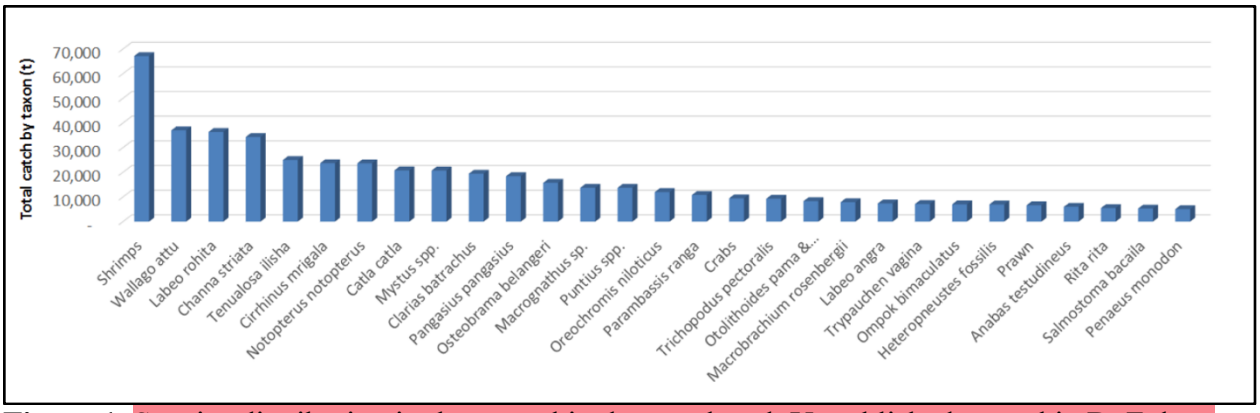


Figure 1. Species distribution in the township data analysed: Unpublished township DoF data, 2015-2016 season. Species taxonomy reflects the original data. *Adapted from Joffre et al. (2018), State of Basin Assessment (SOBA) – Aquaculture in the Ayeyarwady Basin* [3]

Empirical ecotoxicological research of freshwater areas has identified heightened levels of lead (Pb), cadmium (Cd), and arsenic (As) in measures higher than maximum permissible limits (MPL) established by international conventions [6].

As *C. striata* populations often occupy shallower habitats, near to agricultural zones, it is exposed to chemical residues and waste washed down from rural infrastructure [4]. Further agricultural investigations on rainfed and irrigated ecosystems indicate that trace heavy metal concentrations percolate to surrounding water and soil matrices, which compounds bioaccumulation in the fauna inhabiting the freshwater regions [5].

Research Questions

This research aims to answer these five formulated questions through the methodology process.

1. How substantially do heavy metal contents (Pb, Cd, As, Hg, and Cr) in water and *C. striata* gill tissues differ between industrial zones to local rural habitats in the Ayeyarwady region?

2. What are the Bioaccumulation Factors of heavy metals (Pb, Cd, As, Hg, and Cr) in *C. striata*, and which specific heavy metal exhibits the highest level of propensity for bioaccumulation in *C. striata*?

3. How do Bioaccumulation Factor of heavy metals (Pb, Cd, As, Hg, and Cr) relate to the Condition Index of *C. striata* populations in different mediums of industrialized zones and rural habitats?

4. How do concentrations of heavy metals (Pb, Cd, As, Hg, and Cr) in gill tissues of *C. striata* from various impact gradients conform to national and international food safety thresholds for human consumption?

5. Based on observed Bioaccumulation Factor measures and heavy metals concentrations, what are the ecological and economic risks suggested by anthropogenic activities to the *C. striata* aquaculture industry in the Ayeyarwady basin?

Methodology

This research will be a quantitative, field-to-laboratory comparative study, designed to address the empirical gap concerning the extent of anthropogenically introduced heavy metal contamination and its physiological effects on local and commercial *C. striata* populations, and possible implications for safety conventions.

1. **Site Stratification:** Spatial bases for sampling will be determined, establishing a gradient of anthropogenic pressure across six distinctive sites, three of which are industrialized areas – Station IND-01 (in the Pathein Industrial Project), Station IND-02 (in the Hinthada Industrial Zone), and Station IND-03 (in the Myaungmya Industrial Zone)– and three of which are local habitats – Station REF-01 (in the Maubin rural wetlands), Station REF-02 (in the Bogale Township's inland freshwater wetland matrix), and Station REF-03 (in the Pyapon District boundary zone). The stratification of individual sites will facilitate the further steps, initiating natural ecological control environments for the empirical analysis of xenobiotic bioaccumulation in *C. striata* across juxtaposed conditions.

2. **Fish Sample Collection:** A representative sampling of n=20 mature *C. striata* specimens will be collected from each established site, preferably from fisheries or aquaculture stations, with a total of N=120 specimens. Each specimen will be selected in terms of comparable allometry, uniform age range and reproductive sex to curtail confounding variables. At the moment of collection, each fish's body weight (W, in g) and total length (L, in cm) will be collected to fulfill the metrics for the calculation of the condition index (K). Tissue samples from the dorsal muscles will be surgically excised using sterilized titanium instruments, cryogenically stored in vials under standard conventions for atomic absorption spectrometry (AAS) procedures, which will be used to quantify the concentration of heavy metals (Pb, Cd, As, Hg, and Cr) in *C. striata* samples.

3. **Water Sample Collection:** Triplicate sampling of water (1.0 L each) from subsurface depths will be collected from each established site, using acid-washed HDPE vials. Immediately after capture, the vials of water will be rinsed with the ambient water to ensure that it remains conditionally replicable to site water. Following this, the samples will be acidified in situ to a pH of less than 2 to prevent the precipitation of the target heavy metals. The samples will then be contained inside cryogenic chambers, maintained at around 4°C, to be transported to the laboratory for atomic absorption spectrometry (AAS) analysis of heavy metal concentrations (Pb, Cd, As, Hg, and Cr) dissolved in the water specimens.

4. **Heavy Metal Analysis:** Dorsal muscle tissues from the sampled fish will be collected and rinsed with deionized water, then oven-dried in 60-80°C conditions, and finally ground into a fine powder using a mortar and pestle to go through the acid digestion

process through a concentrated analytical-grade mixture of nitric acid (HNO_3) and perchloric acid ($HClO_4$) in a temperature-controlled environment. Synchronously, the sampled water specimens will go through a filtering process in which each specimen is filtered through 0.45-µm membrane filters to remove suspended particulates. The concentration of lead (Pb), cadmium (Cd), arsenic (As), mercury (Hg), and chromium (Cr) in both groups of samples – the digested fish tissues and filtered water specimens – will be quantified using atomic absorption spectrometry (AAS). The final data of heavy metal concentrations will be expressed in milligrams per kilogram dry weight (mg/kg dry weight) for *C. striata* specimens and in milligrams per liter (mg/L) for water specimens.

5. **Data Calculation and Representation:** With the data acquired from atomic absorption spectrometry (AAS) of digested fish tissues and filtered water samples along with length and width measurements of fish specimen, data will be quantified using the bioaccumulation factor (BAF) – where it is calculated as $BAF = C_{fish} / C_{water}$ (in which C_{fish} is the concentration of heavy metals in fish samples, and C_{water} is the concentration of heavy metals in water samples)– and Fulton's condition index (K) – where it is calculated as $K = (W/L^3) \cdot 100$ (in which W is the width of the fish specimen, and L is the length of the fish specimen). From the quantified data, the metrics will be visualized by two graphical frameworks to be interpreted. Firstly, a dual-bar chart plotting the target heavy metals (Pb, Cd, As, Hg, and Cr) in the X-axis against the mean BAF on the Y-axis, with paired bars contrasting the industrial stations (Station IND-01, Station IND-02, and Station IND-03) versus the control sites (Station REF-01, Station REF-02, and Station REF-03) will be graphed. Following this, a scatter plot graph mapping mean BAF values on the X-axis against the condition index (K) on the Y-axis to picture potential relationships between heavy metal contamination and physiological impacts across the sampled populations of *C. striata* will be graphed.

Potential Limitations and Ethical Concerns

This study is limited to one sampling season, meaning seasonal hydrological shifts may influence heavy metal concentrations [10] in the stations in ways of non-anthropogenic actions, such as monsoon dilution effects. Live *C. Striata* specimens will require careful consideration for ethical measure, and each fish will be humanely sedated using clinically approved anesthesia, prior to instantaneous and humane euthanasia to minimize stress during tissue sampling. Laboratory acid digestion will follow standard protocols for waste management and safety to minimize environmental impact.

Expected Outcomes

This research aims to establish a clear relationship between the bioaccumulation factor (BAF) value and the condition index (K) of a fish, with a clear variation. The impact of heavy metal pollutants in *C. striata* habitats on fish well-being is expected to be detrimental, as bioaccumulation is associated as a negative variation to the physiological conditions of fish species. Henceforth, this study is also expected to clarify the impact of anthropogenic activities, such as industrial and commercial actions, on bioaccumulation in fish specimens, predominantly

highlighting an explicit propensity of bioaccumulation in fish samples living inside the zone gradients.

Conclusion

This proposal aims to fill in empirical gaps on ecological health of *C. striata* populations and further link those impacts to issues of public health and national economics of Myanmar. Bridging this gap may help provide insight into the management of infrastructure, agriculture and aquaculture in the Ayeyarwady region, where xenobiotic contamination and pollution are evident. Results of this study may assist in maintaining global safety thresholds for food and health in human populations, native and foreign to the Ayeyarwady environment.

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